

Biology · Harder paper

Real VCAA questions ranked by the published state results. Editable: type your answers straight into this PDF, save it, then mark yourself with the guide at the back.

Name _____ Date _____

117 marks

reading 15 min · writing 150 min

State average on these questions

56 / 117 (48%)

a student at the state average on every question

How to use this paper

1. Open each question with its blue link. It opens the official VCAA exam PDF at the right page. The question text isn't reproduced here because it is VCAA copyright (some exams include third-party material).
2. Answer in this file: click a bubble for multiple choice, and type in the boxes for written answers (long answers scroll inside the box). Methods and science working can be typed in plain text, e.g. $v = \sqrt{2gh} = 9.9 \text{ m/s}$.
3. Keep to the real time limits above, and don't look at the guide until you've finished.
4. Save with Ctrl/Cmd + S. Use a PDF app that supports forms: Adobe Acrobat Reader, Chrome, Edge, Firefox, macOS Preview or Xodo.
5. Mark yourself: in the guide at the back, tick each marking point you earned, enter your marks, and compare your total with the state average.

Why these questions

Every question is a real VCAA exam question, chosen by ranking all the questions from the listed exams by how Victorian students actually scored on them (percentage correct for multiple choice, state average ÷ marks for written questions). This paper takes the hardest questions.

Sources

- [2022 Biology exam \(PDF\)](#) · [examiners' report](#)
- [2023 Biology exam \(PDF\)](#) · [examiners' report](#)
- [2024 Biology exam \(PDF\)](#) · [examiners' report](#)
- [2025 Biology exam \(PDF\)](#) · [examiners' report](#)

Section A – Multiple-choice questions

Open each question in the official exam PDF (click the blue link), then select A–D. One mark each.

Question 1 · 1 mark

2022 VCAA exam · Section A · Question 1 · [open in the official exam PDF \(p. 2\)](#) ↗

Level of structure shown by the folded 3D shape of myoglobin

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 2 · 1 mark

2022 VCAA exam · Section A · Question 10 · [open in the official exam PDF \(p. 7\)](#) ↗

Identifying inputs and outputs on a light-dependent stage diagram

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 3 · 1 mark

2022 VCAA exam · Section A · Question 12 · [open in the official exam PDF \(p. 8\)](#) ↗

Which option is qualitative data in a plant-growing investigation

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 4 · 1 mark

2022 VCAA exam · Section A · Question 16 · [open in the official exam PDF \(p. 11\)](#) ↗

The final electron acceptor in cellular respiration

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 5 · 1 mark

2022 VCAA exam · Section A · Question 17 · [open in the official exam PDF \(p. 11\)](#) ↗

The incorrect statement about enzymes

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 6 · 1 mark

2022 VCAA exam · Section A · Question 18 · [open in the official exam PDF \(p. 11\)](#) ↗

Which test tube (temperature, oxygen) produces the most lactic acid

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 7 · 1 mark

2022 VCAA exam · Section A · Question 20 · [open in the official exam PDF \(p. 12\)](#) ↗

How beneficial microorganisms inside a plant protect it from pathogens

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 8 · 1 mark

2022 VCAA exam · Section A · Question 21 · [open in the official exam PDF \(p. 13\)](#) ↗

Matching a cell in an inflammation diagram to its role

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 9 · 1 mark

2022 VCAA exam · Section A · Question 23 · [open in the official exam PDF \(p. 14\)](#) ↗

Minimum quarantine for possible measles exposure, using an incubation range

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 10 · 1 mark

2022 VCAA exam · Section A · Question 29 · [open in the official exam PDF \(p. 16\)](#) ↗

Why an island koala population founded by a few individuals has low diversity

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 11 · 1 mark

2022 VCAA exam · Section A · Question 33 · [open in the official exam PDF \(p. 17\)](#) ↗

How the absolute age of a 125-million-year-old fossil is found

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 12 · 1 mark

2022 VCAA exam · Section A · Question 37 · [open in the official exam PDF \(p. 19\)](#) ↗

A reasonable conclusion from homologous forelimbs of humans, crocodiles and horses

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 13 · 1 mark

2022 VCAA exam · Section A · Question 39 · [open in the official exam PDF \(p. 21\)](#) ↗

A valid conclusion from a vertebrate phylogeny with labelled ancestors

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 14 · 1 mark

2023 VCAA exam · Section A · Question 3 · [open in the official exam PDF \(p. 3\)](#) ↗

A true statement about a diagram of gene expression and RNA processing

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 15 · 1 mark

2023 VCAA exam · Section A · Question 4 · [open in the official exam PDF \(p. 3\)](#) ↗

Number of codons needed for a 100-amino-acid polypeptide

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 16 · 1 mark

2023 VCAA exam · Section A · Question 19 · [open in the official exam PDF \(p. 8\)](#) ↗

Why glucose is used faster without oxygen

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 17 · 1 mark

2023 VCAA exam · Section A · Question 22 · [open in the official exam PDF \(p. 9\)](#) ↗

What APCs use to present allergens to T cells

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 18 · 1 mark

2023 VCAA exam · Section A · Question 24 · [open in the official exam PDF \(p. 10\)](#) ↗

A reasonable conclusion from a 9-year allergy-immunotherapy graph

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 19 · 1 mark

2023 VCAA exam · Section A · Question 26 · [open in the official exam PDF \(p. 11\)](#) ↗

A true statement about monoclonal antibody production

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 20 · 1 mark

2023 VCAA exam · Section A · Question 35 · [open in the official exam PDF \(p. 15\)](#) ↗

A change that improves accuracy rather than precision

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 21 · 1 mark

2024 VCAA exam · Section A · Question 3 · [open in the official exam PDF \(p. 3\)](#) ↗

What attenuation of the trp operon prevents

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 22 · 1 mark

2024 VCAA exam · Section A · Question 4 · [open in the official exam PDF \(p. 3\)](#) ↗

Choosing the diagram of a eukaryotic gene

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 23 · 1 mark

2024 VCAA exam · Section A · Question 5 · [open in the official exam PDF \(p. 4\)](#) ↗

Interpreting a diagram of CRISPR-Cas9 immunity in *S. pyogenes*

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 24 · 1 mark

2024 VCAA exam · Section A · Question 8 · [open in the official exam PDF \(p. 6\)](#) ↗

Locating the Krebs cycle on a mitochondrion diagram

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 25 · 1 mark

2024 VCAA exam · Section A · Question 9 · [open in the official exam PDF \(p. 6\)](#) ↗

Relative ATP output at four labelled locations in and around a mitochondrion

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 26 · 1 mark

2024 VCAA exam · Section A · Question 17 · [open in the official exam PDF \(p. 10\)](#) ↗

What causes the heat and swelling in gout inflammation

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 27 · 1 mark

2024 VCAA exam · Section A · Question 19 · [open in the official exam PDF \(p. 11\)](#) ↗

What the complement system does

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 28 · 1 mark

2024 VCAA exam · Section A · Question 20 · [open in the official exam PDF \(p. 11\)](#) ↗

Roles of neutrophils only vs both neutrophils and dendritic cells

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 29 · 1 mark

2024 VCAA exam · Section A · Question 26 · [open in the official exam PDF \(p. 13\)](#) ↗

Roles of helper T cells and cytotoxic T cells from a diagram

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 30 · 1 mark

2024 VCAA exam · Section A · Question 29 · [open in the official exam PDF \(p. 15\)](#) ↗

Why tibia length could still change in randomly bred mice

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 31 · 1 mark

2024 VCAA exam · Section A · Question 32 · [open in the official exam PDF \(p. 16\)](#) ↗

How ~300-million-year-old spiny shark fossils would be dated

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 32 · 1 mark

2025 VCAA exam · Section A · Question 11 · [open in the official exam PDF \(p. 9\)](#) ↗

Identifying a lamb's father from a DNA profile

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 33 · 1 mark

2025 VCAA exam · Section A · Question 15 · [open in the official exam PDF \(p. 10\)](#) ↗

A true statement about ATP yield in cellular respiration

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 34 · 1 mark

2025 VCAA exam · Section A · Question 17 · [open in the official exam PDF \(p. 11\)](#) ↗

Naming stages from a table of inputs and outputs

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 35 · 1 mark

2025 VCAA exam · Section A · Question 19 · [open in the official exam PDF \(p. 11\)](#) ↗

Matching an investigation methodology to a valid photosynthesis measurement

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 36 · 1 mark

2025 VCAA exam · Section A · Question 21 · [open in the official exam PDF \(p. 12\)](#) ↗

Initial effect of a competitive inhibitor of ATP synthase

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 37 · 1 mark

2025 VCAA exam · Section A · Question 25 · [open in the official exam PDF \(p. 14\)](#) ↗

What happens inside swollen lymph nodes during an infection

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 38 · 1 mark

2025 VCAA exam · Section A · Question 27 · [open in the official exam PDF \(p. 15\)](#) ↗

Matching an innate immune molecule to its characteristic and function

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 39 · 1 mark

2025 VCAA exam · Section A · Question 29 · [open in the official exam PDF \(p. 16\)](#) ↗

The most likely day of first antigen exposure from an antibody graph

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 40 · 1 mark

2025 VCAA exam · Section A · Question 30 · [open in the official exam PDF \(p. 16\)](#) ↗

Interpreting a primary and secondary antibody response graph

Your answer: ☐ A ☐ B ☐ C ☐ D

Section B – Short-answer questions

Open each question in the official exam PDF, then type your answer to every part in the boxes. Show working where more than one mark is available.

Question 1 (8 marks)

2022 VCAA exam · Section B · Question 5 · [open in the official exam PDF \(p. 29\)](#) ↗

CRISPR-edited wheat: designing sgRNA, how indels silence a gene, and bioethanol from starch using *Z. mobilis*

a. How the sgRNA was designed and what it does (3 marks)

b. How an insertion of one nucleotide or a deletion of seven could stop gene expression (3 marks)

c. How *Z. mobilis* is used to make bioethanol (2 marks)

Question 2 (6 marks)

2023 VCAA exam · Section B · Question 4 · [open in the official exam PDF \(p. 27\)](#) ↗

Coenzymes in ATP production; why red blood cells metabolise glucose the same way with or without oxygen

a. Roles of coenzymes in forming ATP (3 marks)

b. Why glucose use and lactate output are the same with or without oxygen in red blood cells (3 marks)

Question 3 (6 marks)

2023 VCAA exam · Section B · Question 5 · [open in the official exam PDF \(p. 28\)](#) ↗

Using CRISPR-Cas9 against citrus canker; steam-explosion and fermentation of infected trees for biofuel

a. How CRISPR-Cas9 could produce canker-resistant plants, using the pathway (4 marks)

b.i. Why 250 °C was used (1 mark)

b.ii. Why cool to 35 °C before adding yeast (1 mark)

Question 4 (8 marks)

2023 VCAA exam · Section B · Question 7 · [open in the official exam PDF \(p. 32\)](#) ↗

Critically endangered helmeted honeyeaters: genetic drift, egg-swapping, a consequences-based ethical approach

a. How genetic drift could change genetic diversity over time (3 marks)

b. How swapping eggs maintains diversity in the wild populations (2 marks)

c. Define a consequences-based approach and relate it to the program (3 marks)

Question 5 (4 marks)

2023 VCAA exam · Section B · Question 8 · [open in the official exam PDF \(p. 33\)](#) ↗

Neanderthal DNA in modern African, European and Asian genomes

a. Two ways people with African ancestry could carry Neanderthal DNA (2 marks)

b. Why Europeans and Asians carry more Neanderthal DNA (2 marks)

Question 6 (5 marks)

2024 VCAA exam · Section B · Question 2 · [open in the official exam PDF \(p. 26\)](#) ↗

Extended response: the proteome and how cells with the same genes (gut vs fat cells) end up so different

Describe the proteome and explain how organisms achieve diverse cell types, referring to both cells (5 marks)

Question 7 (4 marks)

2024 VCAA exam · Section B · Question 3 · [open in the official exam PDF \(p. 27\)](#) ↗

Yeast fermentation data: why glucose never runs out, and the effect of adding oxygen on ethanol output

a. Two reasons glucose doesn't fall to zero over 500 hours (2 marks)

b. Effect of adding oxygen on ethanol production, with a reason (2 marks)

Question 8 (11 marks)

2024 VCAA exam · Section B · Question 4 · [open in the official exam PDF \(p. 28\)](#) ↗

Photosynthesis as a pathway; C3 vs C4 carbon fixation vs light intensity; photorespiration

a. Why photosynthesis is a biochemical pathway (3 marks)

b. Use the graph to support the claim that C4 plants have an advantage (3 marks)

c. Why C3 CO₂ fixation rises with light and then plateaus above ~1700 lux (2 marks)

d. How C4 plants reduce photorespiration compared with C3 plants (3 marks)

Question 9 (11 marks)

2025 VCAA exam · Section B · Question 2 · [open in the official exam PDF \(p. 26\)](#) ↗

trp operon: how enzymes catalyse, repression and attenuation from an enzyme-activity graph, model limits, a codon change

a. How enzymes catalyse reactions (2 marks)

b. Explain the low activity (0–20 min) and steady activity (after ~60 min), including how repression and attenuation work together (5 marks)

c. Two limitations of the model simulation (2 marks)

d. A consequence of UGG changing from a stop codon to trp, justified (2 marks)

Question 10 (6 marks)

2025 VCAA exam · Section B · Question 5 · [open in the official exam PDF \(p. 31\)](#) ↗

Meningococcal disease: neutrophil roles; adaptive responses to extracellular vs intracellular pathogens

a. Two roles of neutrophils (2 marks)

b. Contrast the adaptive response to an extracellular vs an intracellular pathogen (4 marks)

Question 11 (8 marks)

2025 VCAA exam · Section B · Question 7 · [open in the official exam PDF \(p. 34\)](#) ↗

Asian honey bees in Australia: founder effect, mutation-driven adaptation, and the genetic impact on native bees

a. What the founder effect is and why it applies here (2 marks)

b. The process behind the 481 DNA differences and how it helped expansion (3 marks)

c. Potential genetic change in native bees under this selection pressure (3 marks)

END OF PAPER

Marking guide follows

Stop here until you have finished the paper under exam conditions.

Marking guide

Section A – Multiple-choice questions

✓	Q	Ans	Why	How Victoria went
☐	1	C	A single polypeptide folded into its overall 3D shape is tertiary structure. Quaternary needs two or more polypeptides. <i>Most popular wrong answer: D (19%).</i>	61% correct A 4% · B 16% · C 61% · D 19%
☐	2	C	The light-dependent stage releases O ₂ , uses ADP + Pi to make ATP, and turns NADP ⁺ into NADPH. <i>Most popular wrong answer: B (26%).</i>	52% correct A 12% · B 26% · C 52% · D 10%
☐	3	C	A list of people's preferences is descriptive, so it is qualitative. Temperatures, hours and rainfall are numbers. <i>Most popular wrong answer: A (15%).</i>	64% correct A 15% · B 10% · C 64% · D 11%
☐	4	B	Oxygen accepts the electrons at the end of the ETC (and forms water). Water is the product, not the acceptor. <i>Most popular wrong answer: C (27%).</i>	46% correct A 18% · B 46% · C 27% · D 9%
☐	5	C	Enzymes lower the activation energy. They don't raise the energy of the reactant molecules. <i>Most popular wrong answer: D (23%).</i>	54% correct A 9% · B 14% · C 54% · D 23%
☐	6	C	Low oxygen means anaerobic respiration, so lactic acid, and the rate is fastest at 37 °C. At 75 °C the enzymes are denatured. <i>Most popular wrong answer: B (20%).</i>	57% correct A 4% · B 20% · C 57% · D 18%
☐	7	A	Plants have no antibodies or third line of defence. Beneficial microbes can release toxins or antimicrobials that kill pathogens. <i>Most popular wrong answer: B (18%).</i>	60% correct A 60% · B 18% · C 15% · D 6%
☐	8	D	Neutrophils are phagocytes that engulf pathogens at the site. Mast cells release histamine, but check that the cell label matches the diagram. <i>Most popular wrong answer: C (25%).</i>	61% correct A 10% · B 3% · C 25% · D 61%
☐	9	B	Use the top of the incubation range (7–21 days), not the average: 21 days. <i>Most popular wrong answer: D (42%).</i>	28% correct A 2% · B 28% · C 29% · D 42%
☐	10	B	A few individuals founding a new population is the founder effect. A bottleneck is a crash within the original population. 29% mixed them up. <i>Most popular wrong answer: A (29%).</i>	64% correct A 29% · B 64% · C 6% · D 1%
☐	11	D	Radioisotope dating of the surrounding volcanic rock. Carbon-14 is far too short-lived, and index fossils give only relative ages. <i>Most popular wrong answer: B (27%).</i>	52% correct A 8% · B 27% · C 13% · D 52%
☐	12	D	Homologous limbs point to a common ancestor. Living species aren't transitional forms of one another. <i>Most popular wrong answer: C (28%).</i>	63% correct A 5% · B 3% · C 28% · D 63%
☐	13	A	Frogs branch off after coelacanths, so they arose more recently. Check each claimed ancestor against the actual node positions. <i>Most popular wrong answer: C (21%).</i>	65% correct A 65% · B 6% · C 21% · D 7%
☐	14	D	The cap added to the 5' end of pre-mRNA is a modified guanine (the methyl-G cap). Rule out the others by counting introns and naming each step. <i>Most popular wrong answer: C (26%).</i>	39% correct A 15% · B 20% · C 26% · D 39%
☐	15	C	One codon per amino acid, so 100 codons. 300 is the number of nucleotides, which is where 39% went. <i>Most popular wrong answer: D (39%).</i>	33% correct A 24% · B 4% · C 33% · D 39%

✓	Q	Ans	Why	How Victoria went
☐	16	D	Anaerobic respiration yields only 2 ATP per glucose (vs ~30–32 aerobic), so the cell must break down more glucose to meet its ATP demand. <i>Most popular wrong answer: B (24%).</i>	61% correct A 6% · B 24% · C 9% · D 61%
☐	17	D	Antigens are presented on major histocompatibility complex (MHC) proteins. IgE sits on mast cells and isn't used for presentation. <i>Most popular wrong answer: C (25%).</i>	53% correct A 15% · B 7% · C 25% · D 53%
☐	18	A	Prescriptions stay reduced relative to control across the whole study, so the effect is long-term and sustained. The steepest early drop doesn't mean "most effective then". <i>Most popular wrong answer: C (38%).</i>	49% correct A 49% · B 2% · C 38% · D 11%
☐	19	A	Each hybridoma is a clone of one B cell, so all its antibodies are identical. The mouse B cells, not the cancer cells, come from the mouse. <i>Most popular wrong answer: D (20%).</i>	60% correct A 60% · B 16% · C 4% · D 20%
☐	20	C	Accuracy is limited by systematic error, so calibrate the equipment. Repeating trials improves precision/reliability, which was the popular trap. <i>Most popular wrong answer: B (38%).</i>	55% correct A 2% · B 38% · C 55% · D 5%
☐	21	B	When trp is high, the leader mRNA forms a terminator loop and transcription stops early, so the trp-synthesis enzymes aren't made. Repressor binding to the operator is a separate mechanism. <i>Most popular wrong answer: C (19%).</i>	56% correct A 13% · B 56% · C 19% · D 12%
☐	22	D	Eukaryotic gene: promoter, then alternating exons and introns, then a terminator. Operators belong to prokaryotic operons. <i>Most popular wrong answer: A (21%).</i>	61% correct A 21% · B 10% · C 8% · D 61%
☐	23	A	Viral DNA fragments are inserted as spacers between the repeats, giving a memory of the virus. The PAM sits next to the target in the viral DNA, not in the CRISPR array.	65% correct A 65% · B 12% · C 14% · D 9%
☐	24	B	The Krebs cycle is in the matrix (location 2). The ETC is on the cristae (inner membrane), which was the popular wrong answer. <i>Most popular wrong answer: A (26%).</i>	64% correct A 26% · B 64% · C 4% · D 6%
☐	25	A	ETC on the cristae: high. Krebs cycle in the matrix: low (2). Glycolysis in the cytosol: low (2). Outer membrane: none. <i>Most popular wrong answer: C (23%).</i>	52% correct A 52% · B 18% · C 23% · D 7%
☐	26	C	Signalling molecules such as histamine and cytokines cause vasodilation and leaky capillaries, which produce heat and swelling. Option B starts true but gives the wrong reason. <i>Most popular wrong answer: B (23%).</i>	58% correct A 10% · B 23% · C 58% · D 9%
☐	27	A	Complement proteins enhance inflammation (for example by attracting phagocytes) and can lyse pathogens. They are innate and aren't activated by helper T cells. <i>Most popular wrong answer: B (22%).</i>	55% correct A 55% · B 22% · C 19% · D 3%
☐	28	B	Both are phagocytes, and neutrophils also release chemicals that kill pathogens. Dendritic cells (not neutrophils) present on MHC II. <i>Most popular wrong answer: C (22%).</i>	59% correct A 8% · B 59% · C 22% · D 11%
☐	29	B	Cytotoxic T cells make infected cells undergo apoptosis. Helper T cells release cytokines, not antibodies. Each T cell recognises one antigen, and infected cells present on MHC I, not MHC II. <i>Most popular wrong answer: D (20%).</i>	65% correct A 10% · B 65% · C 5% · D 20%
☐	30	B	In a small population random chance changes allele frequencies, which is genetic drift. There was no gene flow between cages and no selection. <i>Most popular wrong answer: A (48%).</i>	27% correct A 48% · B 27% · C 6% · D 19%
☐	31	C	Carbon-14 only works up to ~50 000 years, so these were dated relatively by rock-layer position (superposition). <i>Most popular wrong answer: B (21%).</i>	55% correct A 20% · B 21% · C 55% · D 4%

✓	Q	Ans	Why	How Victoria went
☐	32	C	Every lamb band not found in the mother must come from the father. Only father C has all of those bands. <i>Most popular wrong answer: B (27%).</i>	48% correct A 5% · B 27% · C 48% · D 19%
☐	33	B	Theoretical ATP yields (30–32 in the current design) differ from the actual yield because of leaks and transport costs. Root cells still respire using transported sugars. <i>Most popular wrong answer: A (21%).</i>	54% correct A 21% · B 54% · C 15% · D 10%
☐	34	B	Match the inputs and outputs: stage U fits the light-dependent stage (H ₂ O in, O ₂ , ATP, NADPH out) and V fits glycolysis (glucose to pyruvate, net 2 ATP). <i>Most popular wrong answer: A (19%).</i>	63% correct A 19% · B 63% · C 9% · D 8%
☐	35	B	The methodology must suit the variable. A controlled experiment can measure glucose, while fieldwork, literature reviews or models don't fit the other options' stages. <i>Most popular wrong answer: C (42%).</i>	36% correct A 12% · B 36% · C 42% · D 10%
☐	36	D	The inhibitor blocks ADP + Pi binding at the active site, so less ATP is made and ADP builds up. It binds the enzyme's active site, not an allosteric site, and not ATP. <i>Most popular wrong answer: A (43%).</i>	34% correct A 43% · B 9% · C 15% · D 34%
☐	37	B	Lymph nodes are where antigen presentation happens and specific lymphocytes undergo clonal selection and expansion, which causes the swelling. <i>Most popular wrong answer: A (19%).</i>	64% correct A 19% · B 64% · C 9% · D 8%
☐	38	C	Complement proteins circulate inactive and, once activated, can form pores that lyse cells. Histamine isn't an enzyme, interferons don't digest cell walls, and lysozyme isn't from mast cells. <i>Most popular wrong answer: A (26%).</i>	58% correct A 26% · B 10% · C 58% · D 5%
☐	39	A	A primary response has a lag of about 10 days before antibodies rise, so exposure was around day 2, not day 13 when antibodies first appear (69% picked that). <i>Most popular wrong answer: B (69%).</i>	27% correct A 27% · B 69% · C 4% · D 1%
☐	40	D	The second peak (day 33) is the secondary response: memory B cells give a faster, larger antibody response. Plasma cells come from B cells, and cytotoxic T cells don't make antibodies. <i>Most popular wrong answer: A (25%).</i>	63% correct A 25% · B 8% · C 4% · D 63%

Multiple-choice score: _____ / 40

Section B – Short-answer questions

Question 1 (8 marks) · state average 3.8 / 8 (48%)

2022 VCAA exam · Section B · Question 5

a. How the sgRNA was designed and what it does

- ☐ Designed to be complementary to the target sequence in the grain-weight gene
- ☐ sgRNA combines with Cas9
- ☐ Guides Cas9 to the target gene so it cuts there

Answer / notes: Match the sgRNA to the target gene sequence. It binds Cas9 and guides it to cut there.

State average 1.4 / 3 (47%); 24% scored full marks. Mark distribution: 0: 30% · 1: 22% · 2: 25% · 3: 24%

Your mark: _____ / 3

b. How an insertion of one nucleotide or a deletion of seven could stop gene expression

- ☐ Both cause a frameshift (not a multiple of three)
- ☐ Codons downstream change, so the amino acid sequence changes
- ☐ Often an early stop codon, so the protein is truncated or non-functional

Answer / notes: Either indel shifts the reading frame, which scrambles the downstream codons, usually giving an early stop and a non-functional protein.

State average 1.2 / 3 (40%); 17% scored full marks. Mark distribution: 0: 38% · 1: 23% · 2: 22% · 3: 17%

Your mark: _____ / 3

c. How *Z. mobilis* is used to make bioethanol

- ☐ It ferments glucose (from starch breakdown)
- ☐ Anaerobically, producing ethanol (and CO₂)

Answer / notes: Enzymes turn starch into glucose, then *Z. mobilis* ferments it anaerobically to ethanol.

State average 1.2 / 2 (60%); 50% scored full marks. Mark distribution: 0: 31% · 1: 19% · 2: 50%

Your mark: _____ / 2

Question 2 (6 marks) · state average 2.3 / 6 (38%)

2023 VCAA exam · Section B · Question 4

a. Roles of coenzymes in forming ATP

- ☐ Loaded coenzymes (NADH, FADH₂) are produced in glycolysis and the Krebs cycle
- ☐ They carry electrons/H⁺ to the electron transport chain
- ☐ Where the energy is used to join ADP + Pi into ATP (and the coenzymes are unloaded and recycled)

Answer / notes: NAD⁺/FAD pick up electrons and H⁺, deliver them to the ETC, drive ATP synthesis, and return unloaded.

State average 1.3 / 3 (43%); 18% scored full marks. Mark distribution: 0: 31% · 1: 25% · 2: 27% · 3: 18%

Your mark: _____ / 3

b. Why glucose use and lactate output are the same with or without oxygen in red blood cells

- ☐ Red blood cells have no mitochondria
- ☐ So they can't carry out aerobic respiration (no Krebs cycle or ETC)
- ☐ They rely only on anaerobic fermentation (glucose to lactate), whatever the oxygen level

Answer / notes: No mitochondria means no aerobic stages, so it is always glycolysis plus lactate, and oxygen makes no difference. 40% scored zero.

State average 1.0 / 3 (33%); 8% scored full marks. Mark distribution: 0: 40% · 1: 27% · 2: 26% · 3: 8%

Your mark: _____ / 3

Question 3 (6 marks) · state average 2.7 / 6 (45%)

2023 VCAA exam · Section B · Question 5

a. How CRISPR-Cas9 could produce canker-resistant plants, using the pathway

- ☐ sgRNA designed complementary to CsWRKY22 (or CsLOB1)
- ☐ sgRNA guides Cas9 to that gene
- ☐ Cas9 cuts at the target (next to the PAM), knocking out the gene
- ☐ So the protein isn't made / CsLOB1 isn't transcribed, and no canker proteins are produced

Answer / notes: Target a gene in the chain (e.g. CsWRKY22). Knocking it out breaks the pathway, so no canker proteins are made.

State average 1.7 / 4 (42%); 14% scored full marks. Mark distribution: 0: 31% · 1: 20% · 2: 15% · 3: 20% · 4: 14%

Your mark: _____ / 4

b.i. Why 250 °C was used

- ☐ It ruptures the most cells (94%), releasing the most glucose

Answer / notes: The most cells burst (94%), so the most sugars are released.

State average 0.5 / 1 (50%); 50% scored full marks. Mark distribution: 0: 50% · 1: 50%

Your mark: _____ / 1

b.ii. Why cool to 35 °C before adding yeast

- ☐ 35 °C suits the yeast's fermentation enzymes, so yeast can ferment the sugars to ethanol

Answer / notes: 35 °C is near the optimum for the fermentation enzymes. At 250 °C the yeast would be destroyed.

State average 0.5 / 1 (50%); 52% scored full marks. Mark distribution: 0: 48% · 1: 52%

Your mark: _____ / 1

Question 4 (8 marks) · state average 3.8 / 8 (48%)

2023 VCAA exam · Section B · Question 7

a. How genetic drift could change genetic diversity over time

- ☐ Diversity decreases
- ☐ Drift is a random change in allele frequencies, strongest in small populations
- ☐ Alleles can be lost (or fixed) purely by chance

Answer / notes: In small populations chance events remove alleles, so diversity falls.

State average 1.3 / 3 (43%); 20% scored full marks. Mark distribution: 0: 37% · 1: 20% · 2: 23% · 3: 20%

Your mark: _____ / 3

b. How swapping eggs maintains diversity in the wild populations

- ☐ Captive-bred birds carry different alleles
- ☐ Swapping eggs creates gene flow, adding those alleles to the wild gene pools

Answer / notes: Egg swaps act as gene flow, bringing new alleles into each small population.

State average 1.0 / 2 (50%); 28% scored full marks. Mark distribution: 0: 30% · 1: 42% · 2: 28%

Your mark: _____ / 2

c. Define a consequences-based approach and relate it to the program

- ☐ Definition: judge an action by its outcomes, maximising benefits and minimising harms
- ☐ Positive outcomes of the program (e.g. species saved, more diversity)
- ☐ Negative effects weighed (e.g. cost, stress to birds, disease risk)

Answer / notes: Consequences-based means weighing outcomes. Saving the species and diversity outweighs the costs and stress to the birds.

State average 1.5 / 3 (50%); 14% scored full marks. Mark distribution: 0: 18% · 1: 28% · 2: 40% · 3: 14%

Your mark: _____ / 3

Question 5 (4 marks) · state average 2.0 / 4 (50%)

2023 VCAA exam · Section B · Question 8

a. Two ways people with African ancestry could carry Neanderthal DNA

- ☐ Ancestors left Africa, interbred with Neanderthals, then some returned (back-migration)
- ☐ Another (shared DNA inherited from the common ancestor, or interbreeding before the populations separated)

Answer / notes: Back-migration of people who had interbred with Neanderthals, or DNA shared from the common ancestor.

State average 1.0 / 2 (50%); 26% scored full marks. Mark distribution: 0: 23% · 1: 51% · 2: 26%

Your mark: _____ / 2

b. Why Europeans and Asians carry more Neanderthal DNA

- ☐ H. sapiens coexisted with Neanderthals in Europe/Asia for a long time
- ☐ So there were more opportunities for interbreeding there (Neanderthals didn't live in Africa)

Answer / notes: Neanderthals lived in Europe and Asia, so H. sapiens there had far more chances to interbreed.

State average 1.0 / 2 (50%); 33% scored full marks. Mark distribution: 0: 32% · 1: 35% · 2: 33%

Your mark: _____ / 2

Question 6 (5 marks) · state average 2.4 / 5 (48%)

2024 VCAA exam · Section B · Question 2

Describe the proteome and explain how organisms achieve diverse cell types, referring to both cells

- ☐ Proteome: the complete set of proteins produced by an organism (or cell)
- ☐ Genes code for proteins, but different genes are expressed in different cells (gene regulation)
- ☐ Alternative splicing lets one gene produce several proteins
- ☐ Epithelial cells express proteins for absorption (e.g. membrane transport proteins, microvilli)
- ☐ Fat cells express proteins for lipid storage/metabolism, so different proteins give different structure and function

Answer / notes: Define the proteome, then gene regulation, alternative splicing and post-translational changes, and tie each to the gut cell and the fat cell.

State average 2.4 / 5 (48%); 10% scored full marks. Mark distribution: 0: 16% · 1: 18% · 2: 15% · 3: 23% · 4: 18% · 5: 10%

Your mark: _____ / 5

Question 7 (4 marks) · state average 1.8 / 4 (45%)

2024 VCAA exam · Section B · Question 3

a. Two reasons glucose doesn't fall to zero over 500 hours

- ☐ Ethanol builds up and becomes toxic to the yeast (or changes pH), stopping fermentation
- ☐ Another limiting factor (yeast numbers, coenzymes, or the yeast's ATP needs already met)

Answer / notes: Ethanol poisons the yeast before all the glucose is used, and other factors (cell numbers, coenzymes) limit the rate. 64% scored zero.

State average 0.5 / 2 (25%); 9% scored full marks. Mark distribution: 0: 64% · 1: 27% · 2: 9%

Your mark: _____ / 2

b. Effect of adding oxygen on ethanol production, with a reason

- ☐ Less (or no) ethanol is produced
- ☐ Because the yeast switches to aerobic respiration (ethanol comes only from anaerobic fermentation)

Answer / notes: Ethanol output drops because the yeast respire aerobically, turning glucose into CO₂ and water instead.

State average 1.3 / 2 (65%); 56% scored full marks. Mark distribution: 0: 29% · 1: 15% · 2: 56%

Your mark: _____ / 2

Question 8 (11 marks) · state average 5.4 / 11 (49%)

2024 VCAA exam · Section B · Question 4

a. Why photosynthesis is a biochemical pathway

- ☐ A series of reactions / stages
- ☐ The outputs of one step are the inputs of the next
- ☐ Each step is catalysed by specific enzymes (with coenzymes)

Answer / notes: It is a series of enzyme-catalysed steps where each product feeds the next (light-dependent to light-independent).

State average 1.4 / 3 (47%); 17% scored full marks. Mark distribution: 0: 25% · 1: 32% · 2: 26% · 3: 17%

Your mark: _____ / 3

b. Use the graph to support the claim that C4 plants have an advantage

- ☐ Quotes data for both plants (e.g. at 400 lux C3 ≈ 90, C4 ≈ 150 mmol/hr)
- ☐ C4 fixes more CO₂ at every light intensity
- ☐ So C4 makes more glucose and grows faster, outcompeting C3

Answer / notes: Quote paired values, state that C4 is higher throughout, then link to more glucose and faster growth.

State average 1.5 / 3 (50%); 16% scored full marks. Mark distribution: 0: 21% · 1: 30% · 2: 33% · 3: 16%

Your mark: _____ / 3

c. Why C3 CO₂ fixation rises with light and then plateaus above ~1700 lux

- ☐ Rising light speeds up the light-dependent stage, giving more ATP/NADPH for the Calvin cycle
- ☐ Above ~1700 lux another factor limits it (e.g. Rubisco saturated, or CO₂/temperature)

Answer / notes: More light means more ATP/NADPH, so fixation rises until something else (Rubisco, CO₂) becomes limiting.

State average 0.7 / 2 (35%); 13% scored full marks. Mark distribution: 0: 43% · 1: 44% · 2: 13%

Your mark: _____ / 2

d. How C4 plants reduce photorespiration compared with C3 plants

- ☐ C4 plants fix CO₂ first in mesophyll cells (into a C4 acid)
- ☐ It is transported to the bundle sheath and released, giving a high CO₂ concentration around Rubisco
- ☐ So Rubisco binds O₂ far less, reducing photorespiration (especially when stomata close in heat)

Answer / notes: Spatial separation: CO₂ is concentrated in the bundle sheath, so Rubisco sees mostly CO₂, not O₂.

State average 1.8 / 3 (60%); 39% scored full marks. Mark distribution: 0: 22% · 1: 14% · 2: 25% · 3: 39%

Your mark: _____ / 3

Question 9 (11 marks) · state average 4.5 / 11 (41%)

2025 VCAA exam · Section B · Question 2

a. How enzymes catalyse reactions

- ☐ The substrate binds a complementary active site (enzyme–substrate complex)
- ☐ The enzyme lowers the activation energy

Answer / notes: Complementary active site, then E–S complex, then lower E_a . “Speeds up reactions” alone scored zero.

State average 0.8 / 2 (40%); 23% scored full marks. Mark distribution: 0: 44% · 1: 33% · 2: 23%

Your mark: _____ / 2

b. Explain the low activity (0–20 min) and steady activity (after ~60 min), including how repression and attenuation work together

- ☐ 0–20 min: trp present, so enzymes are not needed and activity is low
- ☐ Repression: trp binds the repressor, which binds the operator and blocks RNA polymerase
- ☐ Attenuation: with high trp the ribosome moves quickly through the leader, the terminator loop forms and transcription stops early
- ☐ After the shift (no trp): the repressor is inactive and there is no attenuation, so the structural genes are transcribed and enzymes made
- ☐ Activity levels off once enzyme production reaches a steady rate

Answer / notes: Refer to both time periods. Repression is the main on/off switch, and attenuation fine-tunes it. With trp absent, both are released and the enzymes rise to a plateau.

State average 2.4 / 5 (49%); 14% scored full marks. Mark distribution: 0: 21% · 1: 12% · 2: 17% · 3: 16% · 4: 19% · 5: 14%

Your mark: _____ / 5

c. Two limitations of the model simulation

- ☐ One limitation (e.g. oversimplified / assumes ideal conditions)
- ☐ Another (e.g. ignores real-time variation or other factors such as temperature/pH, or only shows a general trend)

Answer / notes: Models simplify: they assume ideal, constant conditions and can’t capture the real lag and overshoot after the shift.

State average 0.8 / 2 (40%); 19% scored full marks. Mark distribution: 0: 38% · 1: 43% · 2: 19%

Your mark: _____ / 2

d. A consequence of UGG changing from a stop codon to trp, justified

- ☐ Identifies a consequence (proteins made longer / translation continues past the old stop, or new trp-containing proteins)
- ☐ Justifies via the change in amino acid sequence and so protein structure/function (possibly an advantage or disadvantage)

Answer / notes: Where UGG used to end translation, trp is now added, so polypeptides can be longer with new structures and functions. Only 6% got both marks.

State average 0.5 / 2 (24%); 6% scored full marks. Mark distribution: 0: 57% · 1: 38% · 2: 6%

Your mark: _____ / 2

Question 10 (6 marks) · state average 2.7 / 6 (44%)

2025 VCAA exam · Section B · Question 5

a. Two roles of neutrophils

- ☐ Phagocytosis of the bacteria
- ☐ Another (migrate to the site, release cytokines, release antimicrobial chemicals/defensins)

Answer / notes: Engulf the bacteria and release chemicals that kill them.

State average 1.2 / 2 (60%); 44% scored full marks. Mark distribution: 0: 24% · 1: 32% · 2: 44%

Your mark: _____ / 2

b. Contrast the adaptive response to an extracellular vs an intracellular pathogen

- ☐ Extracellular: humoral immunity; intracellular: cell-mediated immunity
- ☐ Extracellular: B cells become plasma cells that make antibodies
- ☐ Intracellular: cytotoxic T cells recognise infected cells via MHC I
- ☐ Outcome: antibodies neutralise or tag free pathogens, while cytotoxic T cells induce apoptosis of infected cells

Answer / notes: Use paired contrasts. Humoral (B cells, antibodies) for pathogens outside cells, cell-mediated (Tc, apoptosis) for pathogens inside them. 41% scored zero.

State average 1.5 / 4 (37%); 13% scored full marks. Mark distribution: 0: 41% · 1: 12% · 2: 16% · 3: 18% · 4: 13%

Your mark: _____ / 4

Question 11 (8 marks) · state average 3.5 / 8 (44%)

2025 VCAA exam · Section B · Question 7

a. What the founder effect is and why it applies here

- ☐ A small group founds a new population with reduced genetic diversity compared with the source
- ☐ Only one queen plus her stored sperm founded the population

Answer / notes: Few founders give a small, unrepresentative gene pool. Here it came from one queen and her stored sperm.

State average 1.0 / 2 (50%); 31% scored full marks. Mark distribution: 0: 31% · 1: 38% · 2: 31%

Your mark: _____ / 2

b. The process behind the 481 DNA differences and how it helped expansion

- ☐ Mutation
- ☐ Mutations created new alleles, increasing genetic diversity
- ☐ Bees with advantageous alleles survived and reproduced more (natural selection), so those alleles became more common

Answer / notes: Mutations gave new variation, and selection favoured better foragers, so the colonies spread.

State average 1.2 / 3 (41%); 9% scored full marks. Mark distribution: 0: 30% · 1: 27% · 2: 34% · 3: 9%

Your mark: _____ / 3

c. Potential genetic change in native bees under this selection pressure

- ☐ The native population may shrink, reducing its gene pool / diversity
- ☐ Native bees less able to compete are selected against
- ☐ Alleles that help competition increase in frequency (and others are lost)

Answer / notes: Competition acts as a selection pressure, so less competitive native bees die out, allele frequencies shift and diversity may fall.

State average 1.3 / 3 (45%); 15% scored full marks. Mark distribution: 0: 27% · 1: 26% · 2: 31% · 3: 15%

Your mark: _____ / 3

Your result

Total: _____ / 117 State average on these questions: **56 / 117 (48%)**

Topics to revise:

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